

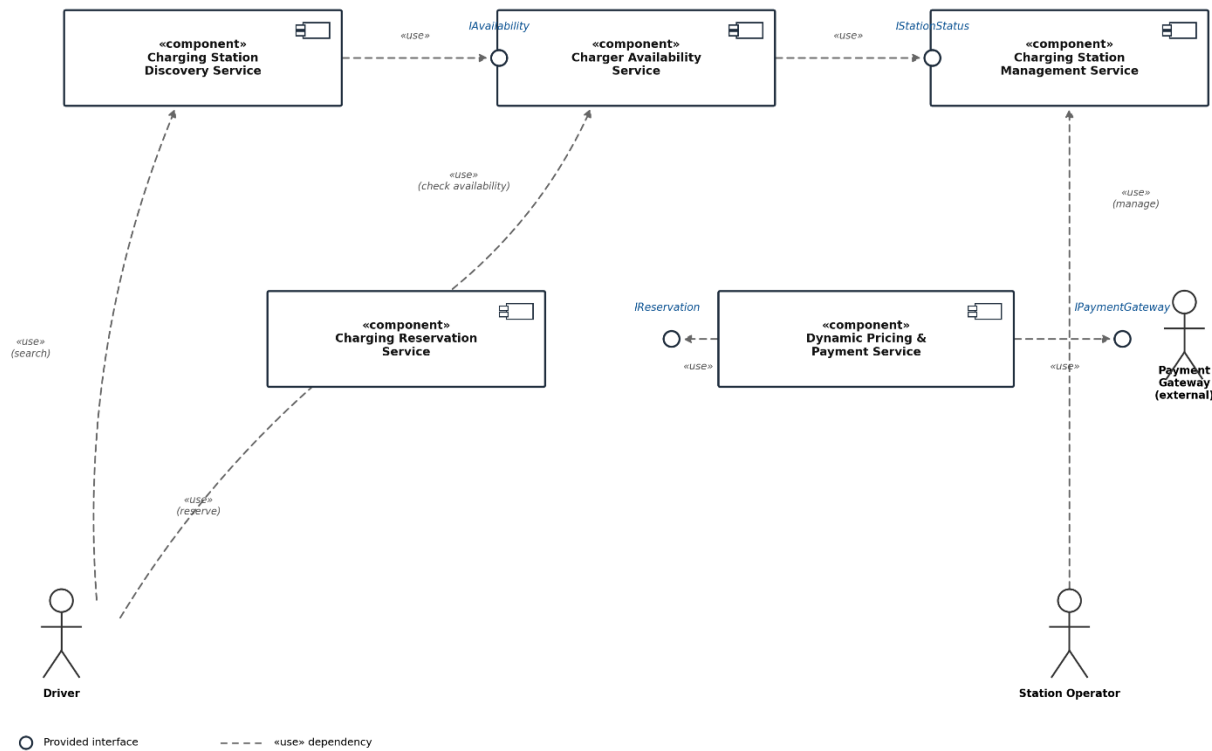
Lab 3: Architectural Pattern Selection

EV Charging Station Reservation System

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EV Charging Station Reservation System — Component Diagram

(Microservices Architecture)



We chose Microservices Architecture for the EV Charging Station Reservation System.

Architectural Choice

The system is decomposed into five independently deployable services, mirroring the epics already defined for this project: Charging Station Discovery Service, Charger Availability Service, Charging Reservation Service, Dynamic Pricing & Payment Service, and Charging Station Management Service. Each service owns a distinct part of the domain and communicates with the others through defined interfaces (IAvailability, IStationStatus, IReservation, and IPaymentGateway to the external payment provider).

Two Reasons for This Choice

1. Independent scaling under uneven load: FR002 and UC-01/UC-02 mean the Discovery and Availability services will receive continuous, high-frequency read traffic as drivers

search for and poll stations, while the Charging Station Management Service is only exercised occasionally by station operators. Microservices let the read-heavy discovery path scale (e.g., extra replicas, caching) independently of the low-traffic management path, which a single monolithic layer could not do without over-provisioning the whole system.

2. Fault isolation around external dependencies: UC-05 depends on a third-party Payment Gateway, an integration point that is more likely to fail or be slow than internal logic. Isolating pricing and payment in their own service means an outage or slowdown there does not block drivers from continuing to search for stations or check charger availability – it only affects the final checkout step, directly supporting NFR001's requirement that unpaid, expired reservations be released automatically rather than leaving the whole system stuck.

Security Advantage

Only the Dynamic Pricing & Payment Service holds credentials for, and communicates with, the external Payment Gateway (IPaymentGateway). The Discovery, Availability, Reservation, and Management services never touch payment credentials or transaction data directly. This narrows the PCI-relevant attack surface to a single, independently secured service instead of spreading card-handling logic across the whole system.

Performance Benefit

Because the Charger Availability Service is a small, frequently-pollled read path (UC-02, NFR003), it can be scaled and cached independently of the heavier, transactional Charging Reservation Service that must enforce exclusive socket locks (FR005, NFR002). This keeps station search and availability checks fast for drivers during peak demand, without the locking logic in the Reservation Service becoming a bottleneck for read traffic, or vice versa.